Motion in a 2 and 3 dimensions

Ch 4 – HRW

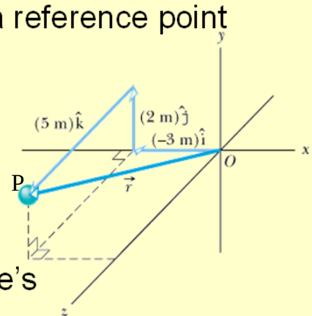
Motion in a plane – 2D
Motion in space – 3D
Projectile motion

Position and Displacement Vectors

- A position vector $\vec{r}$ extends from a reference point (usually the origin *O*) to particle.

  $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$

  $\vec{r} = \left(-3\hat{i} + 2\hat{j} + 5\hat{k}\right)$ m



- If there is a change in the position vector from $\vec{r}_1$ to $\vec{r}_2$ then the particle's displacement is

$$\Delta\vec{r} = \vec{r}_2 - \vec{r}_1 = \left(r_{2x} - r_{1x}\right)\hat{i} + \left(r_{2y} - r_{1y}\right)\hat{j} + \left(r_{2z} - r_{1z}\right)\hat{k}$$
$$= \Delta r_x\hat{i} + \Delta r_y\hat{j} + \Delta r_z\hat{k}$$

- Go through Sample Problem p59.

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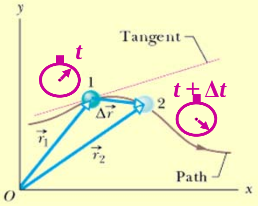
Average and Instantaneous Velocity

- Following the same approach as in Ch 2 we define average velocity:

$$\text{average velocity} = \frac{\text{displacement}}{\text{time}}$$

$$\vec{v}_{avg} = \frac{\Delta\vec{r}}{\Delta t} = \frac{\Delta r_x\hat{i} + \Delta r_y\hat{j} + \Delta r_z\hat{k}}{\Delta t}$$

$$= \frac{\Delta r_x}{\Delta t}\hat{i} + \frac{\Delta r_y}{\Delta t}\hat{j} + \frac{\Delta r_z}{\Delta t}\hat{k}$$



- When speaking of velocity we usually mean the instantaneous velocity given by: $\vec{v} = \frac{d\vec{r}}{dt}$

- Direction of $\vec{v}$ is always tangent (gradient)to the particles path.

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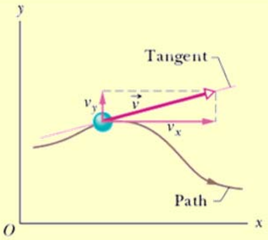
Average and Instantaneous Velocity

- When speaking of velocity we usually mean the instantaneous velocity given by:

$$\vec{v} = \frac{d\vec{r}}{dt} = \frac{d\left(x\hat{i} + y\hat{j} + z\hat{k}\right)}{dt}$$

$$= \frac{dx}{dt}\hat{i} + \frac{dy}{dt}\hat{j} + \frac{dz}{dt}\hat{k}$$

$$= v_x\hat{i} + v_y\hat{j} + v_z\hat{k}$$



- Direction of $\vec{v}$ is always tangent to the particles path.
- Scalar components:

$$v_x = \frac{dx}{dt} \qquad v_y = \frac{dy}{dt} \qquad v_z = \frac{dz}{dt}$$

- Go through Sample problem p62

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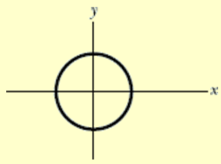
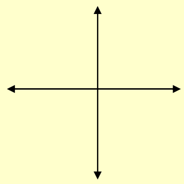
Checkpoint 1

- The figure shows a circular path taken by a particle. If the instantaneous velocity of the particle is $\vec{v} = (2\text{ m/s})\hat{i} - (2\text{ m/s})\hat{j}$ through which quadrant is the particle moving at that instant if it is travelling (a) clockwise and (b) counter-clockwise around the circle? For both cases, draw on the figure.





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Average and Instantaneous Acceleration

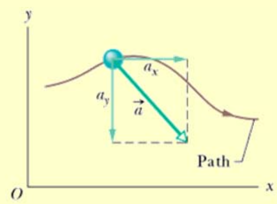
- When a particle's velocity changes in a time interval Δt the average acceleration is:

$$\text{average acceleration} = \frac{\text{change in velocity}}{\text{time interval}}$$

$$\vec{a}_{avg} = \frac{\vec{v}_2 - \vec{v}_1}{\Delta t} = \frac{\Delta\vec{v}}{\Delta t}$$

- As previously discussed the instantaneous acceleration is then:

$$\vec{a} = \frac{d\vec{v}}{dt} = \frac{d}{dt}\left(v_x\hat{i} + v_y\hat{j} + v_z\hat{k}\right)$$
$$= \frac{dv_x}{dt}\hat{i} + \frac{dv_y}{dt}\hat{j} + \frac{dv_z}{dt}\hat{k}$$
$$= a_x\hat{i} + a_y\hat{j} + a_z\hat{k}$$



- Scalar components: $a_x = \frac{dv_x}{dt}$ $a_y = \frac{dv_y}{dt}$ $a_z = \frac{dv_z}{dt}$

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Checkpoint 2

- Here are four descriptions of the position (in meters) of a puck as it moves in an *xy* plane:

1. $x = -3t^2 + 4t - 2$ and $y = 6t^2 - 4t$
2. $x = -3t^3 - 4t$ and $y = -5t^2 + 6$
3. $\vec{r} = 2t^2\hat{i} - (4t+3)\hat{j}$
4. $\vec{r} = (4t^3 - 2t)\hat{i} + 3\hat{j}$

Are the *x* and *y* acceleration components constant? Is the acceleration constant?

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Problem

- A particle with velocity $\vec{v} = -2.0\hat{i} + 4.0\hat{j}$ (in meters per second) at t = 0 undergoes a constant acceleration $\vec{a}$ of magnitude $a = 3.0$ m/s^2 at an angle θ =130° from the positive direction of the x axis. What is the particle's velocity $\vec{v}$ at t = 5 s?
- What do we know? $a = 3.0$ m/s^2 at an angle θ=130° to x axis

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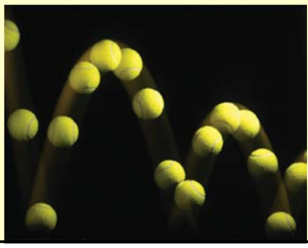
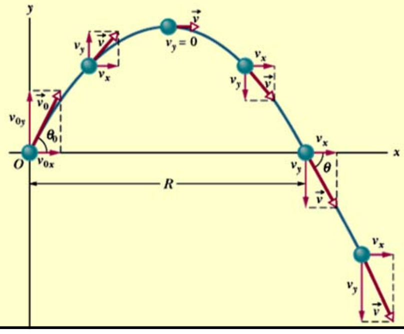
Projectile Motion

- Special case: 2-D motion
- Projectile motion: particle moves in a vertical plane with an initial velocity $\vec{v}_0$ but acceleration is always the freefall of gravity $\vec{a} = -\vec{g} = -9.8$ m/s^2
- Assume no air resistance
- Path is shown
- Initial launch velocity





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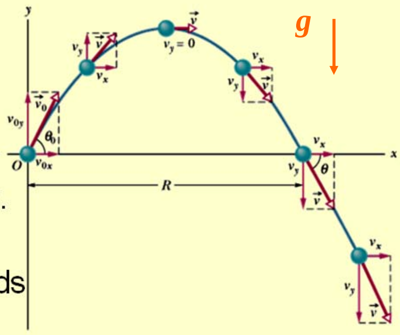
Projectile motion

$$\vec{v}_0 = \vec{v}_{0x}\hat{i} + \vec{v}_{0y}\hat{j}$$
$$v_{0x} = v_0 \cos\theta$$
$$v_{0y} = v_0 \sin\theta$$



- Position vector and velocity vector change continuously.
- Acceleration is ALWAYS constant directed downwards
- NO horizontal acceleration.
- The horizontal motion is independent of the vertical motion.
- Solve the two dimensions separately.

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Projectile Motion Analysed

- Launch point: x_0 and y_0
  - Can be at the origin (but not necessarily)
- Horizontal motion:
  - No acceleration
  - Displacement $x - x_0 = v_{0x}t = (v_0 \cos\theta)t$
- Vertical motion:
  - Acceleration is constant
  - Can use equations from ch 2-9
  - Displacement $y - y_0 = v_{0y}t + ½at^2$

    $= (v_0 \sin\theta)t + ½(-g)t^2$
  - Other equations

    $v_y = v_0 \sin\theta + (-g)t$

    $v_y^2 = (v_0 \sin\theta)^2 + 2(-g)\Delta y$

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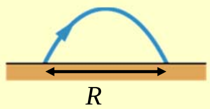
Projectile Motion Analysed

- Horizontal range:
  - When object reaches the ground $y - y_o = 0$
  - There are 2 solutions for x if $y = y_o = 0$

$$y = (\tan\theta_0)x - \frac{gx^2}{2(v_0 \cos\theta_0)^2}$$

  - Horizontal displacement (Range): $x - x_o = R$
  - $R = (v_0 \cos\theta_0)t$
  - and $y = 0 = (v_0 \sin\theta_0)t - gt^2$
  - Eliminate t then

$$R = \frac{v_0^2}{g}\sin 2\theta_0$$



  - Only valid if landing height = launch height
  - Max R when θ = 45° (How do we figure this out?)

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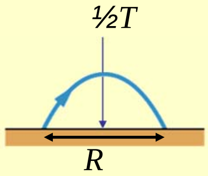
Projectile Motion Analysed

- Time of Flight: (Not in HRW)
  - At max height the time taken is half the total journey.
  - $t = ½T$
  - v = 0 m/s
  - Time of flight T

$$v_y = v_0 \sin\theta - gt$$

$$0 = v_0 \sin\theta - g\left(\frac{T}{2}\right)$$

$$T = \frac{2v_0 \sin\theta}{g}$$



Projectile Motion Analysed

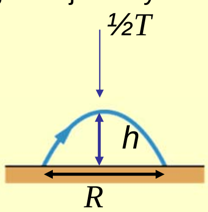
- Max. height: (Not in HRW)
  - At max height is reached halfway through the journey.
  - $t = ½T$
  - v = 0 m/s
  - Max height

$$y - y_0 = (v_0 \sin\theta)t - ½gt^2$$

$$h = (v_0 \sin\theta)\left(\frac{T}{2}\right) - ½g\left(\frac{T}{2}\right)^2$$

$$= (v_0 \sin\theta)\left(\frac{v_0 \sin\theta}{g}\right) - ½g\left(\frac{v_0 \sin\theta}{g}\right)^2$$

$$h = \frac{v_0^2 \sin^2\theta}{2g}$$



- Go through Sample problems p69, p70

Problem

- A rifle with a muzzle velocity v_0 = 450 m/s shoots a bullet at a target 50 m away. How high above the target must the barrel be pointed? Assume the muzzle and the target are at the same height.
- Given:
  - v_0 = 450 m/s
  - R = 50 m
  - g = 9.8 m/s^2

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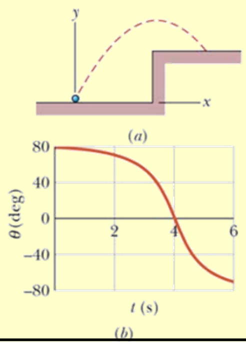
Problem

- At time t = 0 a golf ball is shot from ground level into the air, as shown in the fig. The angle θ between the balls direction of travel and the positive x axis is given as a function of time t. The ball lands at t = 6.00 s.
  - What is the magnitude v_0 of the ball's launch velocity?
  - At what height (y-y_0) above the launch level does the ball land?
  - What is the balls direction of travel just before it lands?



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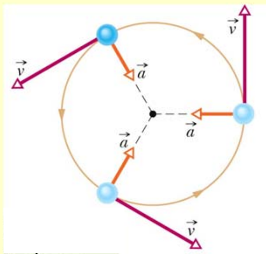
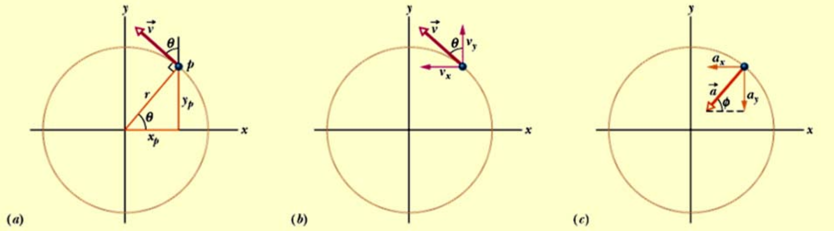
Uniform Circular Motion

- A particle is in uniform circular motion if:
  - It travels in a circular arc
  - At a constant speed.
- Although speed is constant, there is acceleration
  - Velocity is changing direction
  - Velocity is directed tangent to the circle
  - Acceleration is directed to centre of the circle – centripetal acceleration.



- Acceleration $a = \dfrac{v^2}{r}.$
- Period of revolution $T = \dfrac{2\pi r}{v}.$

Uniform Circular Motion-derivation of $a = \dfrac{v^2}{r}$



- Recall - $\vec{v}$ is tangent to path (position) of particle.
- At P : coordinates: x_p, y_p

  position: $\vec{r}$ is at angle θ to x-axis

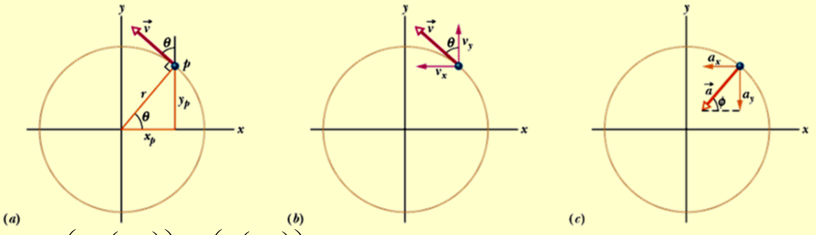
  $\vec{v} \perp$ position

  $\vec{v}$ is at angle θ to perpendicular at P (Fig(b))

  $\vec{v} = v_x\hat{i} + v_y\hat{j} = (-v\sin\theta)\hat{i} + (v\cos\theta)\hat{j}$
- Using Fig (a)

$$\vec{v} = \left(-v\left(\frac{y_p}{r}\right)\right)\hat{i} + \left(v\left(\frac{x_p}{r}\right)\right)\hat{j}$$

Uniform Circular Motion



- $\vec{v} = \left(-v\left(\frac{y_p}{r}\right)\right)\hat{i} + \left(v\left(\frac{x_p}{r}\right)\right)\hat{j}$
- $\vec{a} = \frac{d\vec{v}}{dt} = \left(-\frac{v}{r}\left(\frac{dy_p}{dt}\right)\right)\hat{i} + \left(\frac{v}{r}\left(\frac{dx_p}{dt}\right)\right)\hat{j} = \left(-\frac{v^2}{r}\cos\theta\right)\hat{i} + \left(-\frac{v^2}{r}\sin\theta\right)\hat{j}$
- $a = \sqrt{a_x^2 + a_y^2} = \frac{v^2}{r}\sqrt{\cos^2\theta + \sin^2\theta} = \frac{v^2}{r}$
- Direction of acceleration: $\tan\phi = \frac{a_y}{a_x} = \frac{-\frac{v^2}{r}\sin\theta}{\frac{v^2}{r}\cos\theta} = \tan\theta$

  along radius of circle towards centre

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Checkpoint 5

- An object moves at constant speed along a circular path in a horizontal xy plane, with the centre at the origin. When the object is at x = -2m, its velocity is $-(4\text{m/s})\hat{j}$. Give the object's (a) velocity and (b) acceleration at y = 2 m.

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Problem

- A satellite is at an altitude h = 200 km above the earth (g = 9.2 m.s^2). Determine the satellite's velocity and its period.
  The earth's radius is 6.37×10^6 m.

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Relative Motion – 1-D

- Suppose you watch a passenger on a bus travelling east at 40 km/h. To a passenger on another bus, travelling next to the first at the same velocity, the first passenger seems to be stationary.
- Velocity of particle depends on reference frame of the observer.
- Reference frame is a physical object to which we attach coordinate system.

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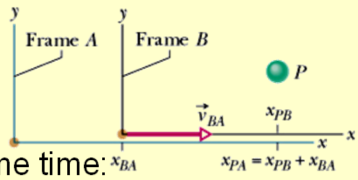
Relative Motion – 1-D

- Suppose Alex is parked at side of road watching car P.
- Bob also watches car P from his car as he moves along the road at constant speed.
- If both Alex and Bob measure the position of car P at the same time:



$$x_{PA} = x_{PB} + x_{BA}$$

- Determining the velocities (derivative):

$$\frac{dx_{PA}}{dt} = \frac{dx_{PB}}{dt} + \frac{dx_{BA}}{dt}$$

$$v_{PA} = v_{PB} + v_{BA}$$

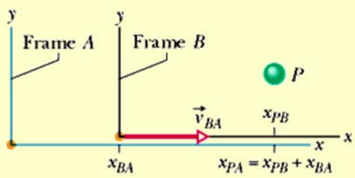
- Velocity components:
  - The velocity of P as measured by A =the velocity of P as measured by B plus the velocity of B as measured by A.

Relative Motion – 1-D

- Determine acceleration:

$$\frac{dv_{PA}}{dt} = \frac{dv_{PB}}{dt} + \frac{dv_{BA}}{dt}$$

$$a_{PA} = a_{PB} + a_{BA} = a_{PB} + 0$$

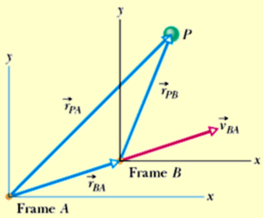


- Observers on different frames of reference that move at constant velocity relative to each other will measure the same acceleration for a moving particle.

Sample Problem p74

- Suppose that Bob's velocity relative to Alex is a constant v_{BA} = 52 km/h and car P is moving in the negative direction of the x axis. (a) If Alex measures a constant v_{PA} = -78 km/h for car P, what velocity v_{PB} will Bob measure? (b) If car P brakes to stop relative to Alex in time t = 10 s at constant acceleration, what is its acceleration a_{PA} relative to Alex? (c) What is the acceleration a_{PB} of car P relative to Bob during braking?

Relative Motion – 2-D



- Two observers watch a particle P from the origins of their reference frames A and B
- B moves at a constant velocity $\vec{v}_{BA}$ relative to A.
- At a particular instant:

$$\vec{r}_{PA} = \vec{r}_{PB} + \vec{r}_{BA}$$
$$\vec{v}_{PA} = \vec{v}_{PB} + \vec{v}_{BA}$$
$$\vec{a}_{PA} = \vec{a}_{PB} + \vec{a}_{BA} = \vec{a}_{PB} + 0$$

Sample Problem p75

- A plane moves due E while the pilot points the plane somewhat S of E, toward a steady wind that blows to the NE. The plane has velocity $\vec{v}_{PW}$ relative to the wind, with an airspeed, relative to the wind of 215 km/h directed at angle θ S of E. The wind has a velocity $\vec{v}_{WG}$ relative to the ground with a speed 65 km/h, directed 20.0° E of N. What is the magnitude of the velocity $\vec{v}_{PG}$ of the plane relative to the ground, and what is θ.

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Problems

- 4. The minute hand on a clock is 10 cm long from tip to support point around which it rotates. Determine the magnitude and angle of the displacement vector of the tip for the following time intervals: (a) quarter past the hour to half past the hour. (b) the next half hour. (c) the next hour after that.
- 8. A plane flies 483 km E from city A to city B in 45.0 min and then 966 km south from city B to city C in 1.50 h. For the whole trip, what are (a) magnitude and direction of the plane's displacement (b) magnitude and direction of average velocity and (e) the plane's average speed?
- 33. A plane, diving with constant speed at at angle of 53° with the vertical, releases a projectile at an altitude of 730 m. The projectile hits the ground 5.00 s after release. (a) What is the speed of the plane? (b) How far does the projectile travel horizontally during its flight? What are the (c) horizontal and (d) vertical components of its velocity just before striking the ground?
- 64. A particle moves horizontally in uniform circular motion, over a horizontal xy plane. At one instant, it moves through the point at coordinates (4.00m, 4.00m) with a velocity of -5.00$\hat{i}$m/s and an acceleration of + 12.5$\hat{j}$ m/s^2. What are the x and y coordinates of the centre of the circular path?

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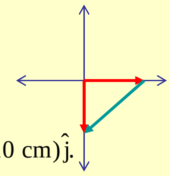
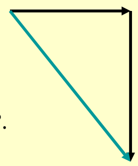
Solutions



- 4(a) $\vec{r}_1 = (10\text{ cm})\hat{\text{i}}$ and $\vec{r}_2 = (-10\text{ cm})\hat{\text{j}}$. $\Delta\vec{r} = \vec{r}_2 - \vec{r}_1 = (-10\text{ cm})\hat{\text{i}} + (-10\text{ cm})\hat{\text{j}}$.

  $|\Delta\vec{r}| = \sqrt{(-10\text{ cm})^2 + (-10\text{ cm})^2} = 14\text{ cm}$. $\theta = \tan^{-1}\left(\dfrac{-10\text{ cm}}{-10\text{ cm}}\right) = 45°$ or $-135°$.

- 4(b) Δr = 20 cm, IΔrI = 20 cm θ = 90°
- 4(c) Δr = 0 cm, IΔrI = 0 cm θ = 0°



- 8(a) $\vec{r}_{AC} = \vec{r}_{AB} + \vec{r}_{BC} = (483\text{ km})\hat{\text{i}} - (966\text{ km})\hat{\text{j}}$

  $|\vec{r}_{AC}| = \sqrt{(483\text{ km})^2 + (-966\text{ km})^2} = 1.08\times10^3\text{ km}$.

  $\theta = \tan^{-1}\left(\dfrac{-966\text{ km}}{483\text{ km}}\right) = -63.4°$.

- 8(b) $\vec{v}_{\text{avg}} = \dfrac{(483\text{ km})\hat{\text{i}} - (966\text{ km})\hat{\text{j}}}{2.25\text{ h}} = (215\text{ km/h})\hat{\text{i}} - (429\text{ km/h})\hat{\text{j}}$.

  $|\vec{v}_{\text{avg}}| = \sqrt{(215\text{ km/h})^2 + (-429\text{ km/h})^2} = 480\text{ km/h}$.

  26.6° east of south

- 8(c) $\text{speed} = \dfrac{\text{distance}}{\text{time}} = \dfrac{483\text{ km} + 966\text{ km}}{2.25\text{ h}} = 644\text{ km/h}$.

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